

ATIVIDADE 1 : INFORMATION ON A LINUX MACHINE

SYSTEM INFORMATION

Estabelecendo conexão com a vpn, através do arquivo .ovpn:

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~]
$ cd Desktop
(kali@kali)-[~/Desktop]
$ sudo openvpn academy-regular\ \((1\).ovpn
[sudo] password for kali:
2026-05-04 10:19:36 DEPRECATED OPTION: --persist-key option ignored. Keys are
now always persisted across restarts.
2026-05-04 10:19:36 WARNING: Compression for receiving enabled. Compression h
as been used in the past to break encryption. Compression support is deprecate
d and we recommend to disable it completely.
2026-05-04 10:19:36 Note: --data-ciphers-fallback with cipher 'AES-128-CBC' d
isables data channel offload.
2026-05-04 10:19:36 OpenVPN 2.7.1 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [
LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
2026-05-04 10:19:36 library versions: OpenSSL 3.6.1 27 Jan 2026, LZO 2.10
2026-05-04 10:19:36 DCO version: N/A
2026-05-04 10:19:37 TCP/UDP: Preserving recently used remote address: [AF_INE
T]38.46.224.172:443
2026-05-04 10:19:37 Socket Buffers: R=[131072→131072] S=[16384→16384]
2026-05-04 10:19:37 Attempting to establish TCP connection with [AF_INET]38.4
6.224.172:443
2026-05-04 10:19:37 TCP connection established with [AF_INET]38.46.224.172:44
3
2026-05-04 10:19:37 TCPv4_CLIENT link local: (not bound)
2026-05-04 10:19:37 TCPv4_CLIENT link remote: [AF_INET]38.46.224.172:443
2026-05-04 10:19:37 TLS: Initial packet from [AF_INET]38.46.224.172:443, sid=
38d89c85 6be7d79b
```

Comando IP A para saber o meu endereço ip dentro da rede:

```
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 84452sec preferred_lft 84452sec
    inet6 fd17:625c:f037:2:6120:479f:f948:ee1f/64 scope global dynamic nopref
ixroute
        valid_lft 86345sec preferred_lft 14345sec
    inet6 fe80::70b2:42f5:23e2:dd3c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel st
ate UNKNOWN group default qlen 500
    link/none
    inet 10.10.16.119/23 brd 10.10.17.255 scope global tun0
        valid_lft forever preferred_lft forever
    inet6 dead:beef:4::1075/64 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::69fb:5870:4c25:fb2d/64 scope link stable-privacy proto kernel
```

Pingando o alvo:

```
(kali㉿kali)-[~]  
$ ping 10.129.86.151  
PING 10.129.86.151 (10.129.86.151) 56(84) bytes of data.  
64 bytes from 10.129.86.151: icmp_seq=1 ttl=63 time=120 ms  
64 bytes from 10.129.86.151: icmp_seq=2 ttl=63 time=121 ms  
64 bytes from 10.129.86.151: icmp_seq=3 ttl=63 time=121 ms  
64 bytes from 10.129.86.151: icmp_seq=4 ttl=63 time=121 ms  
64 bytes from 10.129.86.151: icmp_seq=5 ttl=63 time=120 ms  
^C  
— 10.129.86.151 ping statistics —  
5 packets transmitted, 5 received, 0% packet loss, time 4022ms  
rtt min/avg/max/mdev = 120.012/120.584/120.966/0.371 ms
```

Conectando na máquina-alvo via SSH.

O SSH (Secure Shell) é o protocolo que permite que você opere o terminal de uma máquina remota como se estivesse sentado na frente dela:

```
(kali㉿kali)-[~]  
$ ssh htb-student@10.129.86.151  
The authenticity of host '10.129.86.151 (10.129.86.151)' can't be established  
ED25519 key fingerprint is: SHA256:PHsjpBEAl6hSCzjVohppUybupbLXdBZy8FqtwlMpmj  
U  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.129.86.151' (ED25519) to the list of known hos  
ts.  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
htb-student@10.129.86.151's password:  
Permission denied, please try again.  
htb-student@10.129.86.151's password:  
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 4.15.0-123-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/advantage
```

- **ssh**: O comando que inicia o protocolo.
- **htb-student**: O nome do usuário que você quer "incorporar" na máquina alvo.
- **@**: O separador que diz "conectar este usuário em...".
- **10.129.86.151**: O endereço IP da máquina alvo.

QUESTÕES:

- 1) Find out the machine hardware name and submit it as the answer:

```
htb-student@nixfund:~$ uname -a
Linux nixfund 4.15.0-123-generic #126-Ubuntu SMP Wed Oct 21 09:40:11 UTC 2020
x86_64 x86_64 x86_64 GNU/Linux
htb-student@nixfund:~$
```

2) What is the path to htb-student's home directory?

```
htb-student@nixfund:~$ pwd
/home/htb-student
```

3) What is the path to the htb-student's mail?

```
htb-student@nixfund:~$ echo $MAIL
/var/mail/htb-student
```

Echo \$MAIL: Exibe o caminho completo do arquivo de "caixa de entrada" (mailbox) do usuário atual. O \$MAIL é uma variável de ambiente. O sistema operacional a utiliza para saber onde deve entregar e armazenar as notificações e e-mails internos destinados ao seu usuário

4) Which shell is specified for the htb-student user?

Para descobrir qual shell está definido para um usuário específico no Linux, você precisa consultar o "banco de dados" de usuários do sistema. O arquivo mestre para isso é o /etc/passwd.

```
htb-student@nixfund:~$ grep ``htb-student`` /etc/passwd
htb-student:x:1002:1002::/home/htb-student:/bin/bash
htb-student@nixfund:~$
```

Which kernel release is installed on the system? (Format: 1.22.3)

```
htb-student@nixfund:~$ uname -r
4.15.0-123-generic
```

What is the name of the network interface that MTU is set to 1500?

```
htb-student@nixfund:~$ ifconfig
ens192: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.129.86.151 netmask 255.255.0.0 broadcast 10.129.255.255
    inet6 dead:beef::a0de:adff:feac:a569 prefixlen 64 scopeid 0x0<global>
l>
    inet6 fe80::a0de:adff:feac:a569 prefixlen 64 scopeid 0x20<link>
    ether a2:de:ad:ac:a5:69 txqueuelen 1000 (Ethernet)
    RX packets 8653 bytes 684637 (684.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2092 bytes 183050 (183.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```